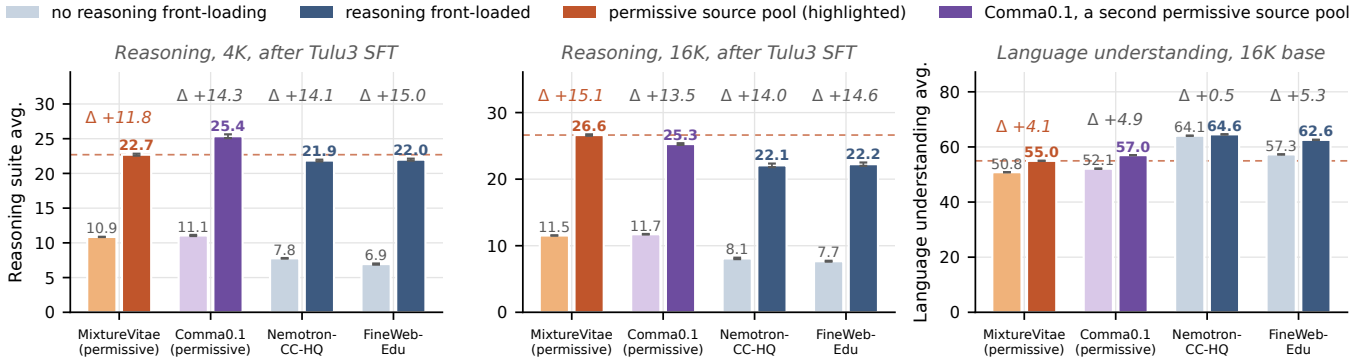
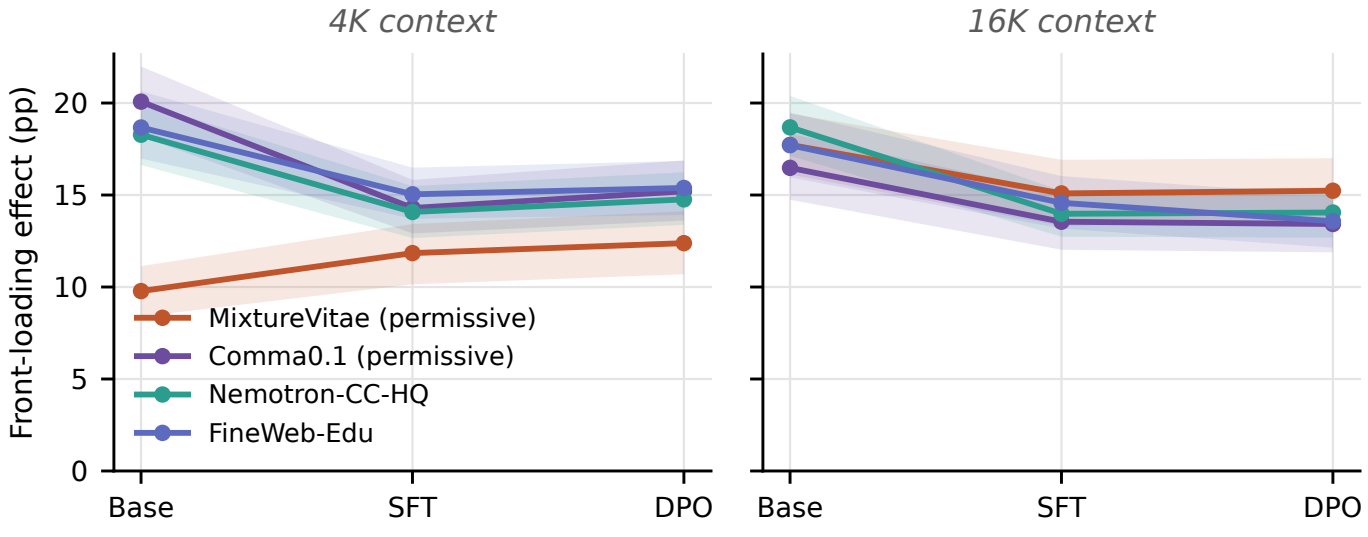
Table 1: **Front-loading versus source licensing at 300B pretraining tokens.** Each *no* and *yes* row is one 1.7B model trained under the identical `open-sci-ref` configuration, differing only in the source pool and in whether the MixtureVitae instruction-and-reasoning subset is included. Reasoning columns are the eleven-benchmark suite average under Protocol A (Appendix Table **??**). Values are mean$_{\pm SD}$ over decoding seeds 42, 1234, and 2025. Language columns are the eleven-task understanding suite on the corresponding base checkpoint, single seed. Δ rows give the front-loading effect in percentage points. Comma0.1 is a second permissive source pool. Its model without front-loading is the Comma0.1 model of Table **??**, and the front-loaded MixtureVitae model is the main MixtureVitae model. Both are reused checkpoints, trained earlier than the other models here with the same recorded `open-sci-ref` configuration. Table **??** gives 95% intervals for the language contrasts.

		Reasoning, 4K			Reasoning, 16K			Language	
Source pool	**Front-loaded**	Base	SFT	DPO	Base	SFT	DPO	4K	16K
MixtureVitae *(permissive)*	no	$7.3_{\pm 0.03}$	$10.9_{\pm 0.02}$	$11.2_{\pm 0.08}$	$11.3_{\pm 0.03}$	$11.5_{\pm 0.02}$	$11.7_{\pm 0.13}$	51.0	50.8
	yes	$17.0_{\pm 0.38}$	$22.7_{\pm 0.14}$	$23.6_{\pm 0.17}$	$29.1_{\pm 0.69}$	$26.6_{\pm 0.11}$	$26.9_{\pm 0.04}$	55.8	55.0
	Δ	+9.8	+11.8	+12.4	+17.7	+15.1	+15.2	+4.8	+4.1
Comma0.1 *(permissive)*	no	$7.3_{\pm 0.12}$	$11.1_{\pm 0.08}$	$11.5_{\pm 0.07}$	$11.1_{\pm 0.22}$	$11.7_{\pm 0.04}$	$12.0_{\pm 0.04}$	50.0	52.1
	yes	$27.4_{\pm 0.14}$	$25.4_{\pm 0.27}$	$26.7_{\pm 0.30}$	$27.5_{\pm 0.34}$	$25.3_{\pm 0.13}$	$25.4_{\pm 0.27}$	56.6	57.0
	Δ	+20.1	+14.3	+15.2	+16.5	+13.5	+13.4	+6.6	+4.9
Nemotron-CC-HQ	no	$3.8_{\pm 0.02}$	$7.8_{\pm 0.04}$	$8.0_{\pm 0.07}$	$7.8_{\pm 0.07}$	$8.1_{\pm 0.11}$	$8.3_{\pm 0.06}$	64.2	64.1
	yes	$22.1_{\pm 0.22}$	$21.9_{\pm 0.13}$	$22.8_{\pm 0.25}$	$26.5_{\pm 0.39}$	$22.1_{\pm 0.27}$	$22.4_{\pm 0.26}$	64.6	64.6
	Δ	+18.3	+14.1	+14.8	+18.7	+14.0	+14.0	+0.4	+0.5
FineWeb-Edu	no	$5.7_{\pm 0.05}$	$6.9_{\pm 0.08}$	$7.0_{\pm 0.06}$	$8.3_{\pm 0.10}$	$7.7_{\pm 0.07}$	$8.2_{\pm 0.07}$	56.7	57.3
	yes	$24.4_{\pm 0.05}$	$22.0_{\pm 0.14}$	$22.4_{\pm 0.08}$	$26.0_{\pm 0.06}$	$22.2_{\pm 0.25}$	$21.8_{\pm 0.07}$	62.8	62.6
	Δ	+18.7	+15.0	+15.4	+17.7	+14.6	+13.6	+6.1	+5.3



factorial_main.pdf



frontloading_trajectory.pdf

Table 2: All Protocol A contrasts on the eleven-benchmark suite average, with 95% paired item-level bootstrap intervals (2,000 replicates, items resampled within benchmark, decoding seeds averaged first) and Benjamini–Hochberg adjusted q-values computed within each family. Positive values favour front-loading in the first block and the permissive source pool in the second. Comma0.1 is a second permissive source pool, so the permissiveness and interaction families include its contrasts, and each family's q-values are adjusted over all of them. Table **??** gives the same contrasts on the language suite.

		4K context			16K context		
Contrast	**Stage**	**Estimate**	**95% CI**	q	**Estimate**	**95% CI**	q
Front-loading effect, within source pool							
MixtureVitae	Base	+9.78	[+8.39, +11.13]	< 0.001	+17.73	[+16.11, +19.42]	< 0.001
	SFT	+11.84	[+10.14, +13.43]	< 0.001	+15.09	[+13.32, +16.92]	< 0.001
	DPO	+12.38	[+10.69, +14.12]	< 0.001	+15.23	[+13.54, +17.00]	< 0.001
Comma0.1	Base	+20.08	[+18.17, +21.99]	< 0.001	+16.48	[+14.74, +18.33]	< 0.001
	SFT	+14.29	[+12.91, +15.82]	< 0.001	+13.55	[+12.02, +15.12]	< 0.001
	DPO	+15.20	[+13.61, +16.89]	< 0.001	+13.42	[+11.88, +14.96]	< 0.001
Nemotron-CC-HQ	Base	+18.28	[+16.63, +20.02]	< 0.001	+18.68	[+17.14, +20.39]	< 0.001
	SFT	+14.08	[+12.66, +15.48]	< 0.001	+13.98	[+12.73, +15.23]	< 0.001
	DPO	+14.76	[+13.38, +16.24]	< 0.001	+14.05	[+12.68, +15.42]	< 0.001
FineWeb-Edu	Base	+18.67	[+16.98, +20.64]	< 0.001	+17.71	[+15.96, +19.49]	< 0.001
	SFT	+15.04	[+13.71, +16.49]	< 0.001	+14.58	[+13.18, +16.03]	< 0.001
	DPO	+15.38	[+13.91, +16.85]	< 0.001	+13.58	[+12.14, +14.99]	< 0.001
Permissive minus non-permissive source pool, both front-loaded							
MV − Nemotron	Base	−5.02	[−6.92, −3.02]	< 0.001	+2.58	[+0.85, +4.27]	0.009
	SFT	+0.84	[−0.96, +2.55]	0.404	+4.55	[+2.80, +6.43]	< 0.001
	DPO	+0.83	[−0.79, +2.50]	0.355	+4.53	[+2.57, +6.37]	< 0.001
MV − FineWeb	Base	−7.35	[−9.27, −5.49]	< 0.001	+3.01	[+1.12, +4.86]	0.005
	SFT	+0.71	[−0.89, +2.29]	0.415	+4.38	[+2.57, +6.23]	< 0.001
	DPO	+1.26	[−0.56, +3.13]	0.223	+5.11	[+3.44, +6.86]	< 0.001
Comma − Nemotron	Base	+5.36	[+3.58, +7.20]	< 0.001	+1.05	[−0.74, +2.96]	0.303
	SFT	+3.50	[+1.84, +5.26]	< 0.001	+3.20	[+1.73, +4.67]	< 0.001
	DPO	+3.89	[+2.26, +5.58]	< 0.001	+3.00	[+1.50, +4.41]	< 0.001
Comma − FineWeb	Base	+3.03	[+1.33, +4.79]	< 0.001	+1.49	[−0.20, +3.21]	0.098
	SFT	+3.37	[+1.69, +5.05]	< 0.001	+3.02	[+1.55, +4.49]	< 0.001
	DPO	+4.32	[+2.66, +6.10]	< 0.001	+3.58	[+2.20, +5.03]	< 0.001
Interaction: front-loading effect, permissive minus non-permissive							
MV − non-permissive	Base	−8.70	[−10.56, −6.88]	0.003	−0.47	[−2.40, +1.33]	0.799
	SFT	−2.72	[−4.55, −1.06]	0.003	+0.81	[−1.11, +2.84]	0.664
	DPO	−2.69	[−4.57, −0.74]	0.034	+1.42	[−0.63, +3.47]	0.329
Comma − non-permissive	Base	+1.60	[−0.06, +3.35]	0.150	−1.72	[−3.59, +0.08]	0.150
	SFT	−0.26	[−1.91, +1.44]	0.829	−0.73	[−2.41, +0.91]	0.621
	DPO	+0.13	[−1.60, +1.94]	0.855	−0.39	[−2.08, +1.30]	0.799

Table 3: Robustness of the two headline quantities to how the suite average is weighted, at 16K context after Tulu3 SFT. The eleven-task macro average implicitly weights math at 45% and code at 27%, and under every view below each front-loading effect and each source-pool contrast stays positive. Protocol A (Appendix Table **??**). Each cell is a point estimate, the difference of two scores that are each averaged over decoding seeds 42, 1234, and 2025. Comma0.1 is a second permissive source pool. Its model without front-loading is the Comma0.1 model of Table **??**, and the front-loaded MixtureVitae model is the main MixtureVitae model. Both are reused checkpoints, trained earlier than the other models here with the same recorded `open-sci-ref` configuration.

	Front-loading effect				**Permissive − non-permissive**			
Averaging view	MV	Comma	Nemotron	FineWeb	MV vs. Nemotron	MV vs. FineWeb	Comma vs. Nemotron	Comma vs. FineWeb
Suite average (11 tasks)	+15.1	+13.5	+14.0	+14.6	+4.6	+4.4	+3.2	+3.0
Equal-domain average	+11.6	+10.9	+12.8	+13.4	+2.6	+3.0	+2.3	+2.7
Drop math domain	+10.8	+11.5	+15.1	+15.2	+2.7	+3.5	+3.0	+3.7
Drop code domain	+13.5	+11.2	+10.3	+11.0	+3.6	+3.0	+2.1	+1.5
Drop science domain	+18.4	+16.2	+16.1	+17.1	+5.9	+5.8	+4.0	+3.8
Drop instruction domain	+15.9	+14.3	+14.3	+14.8	+5.2	+4.8	+3.5	+3.1

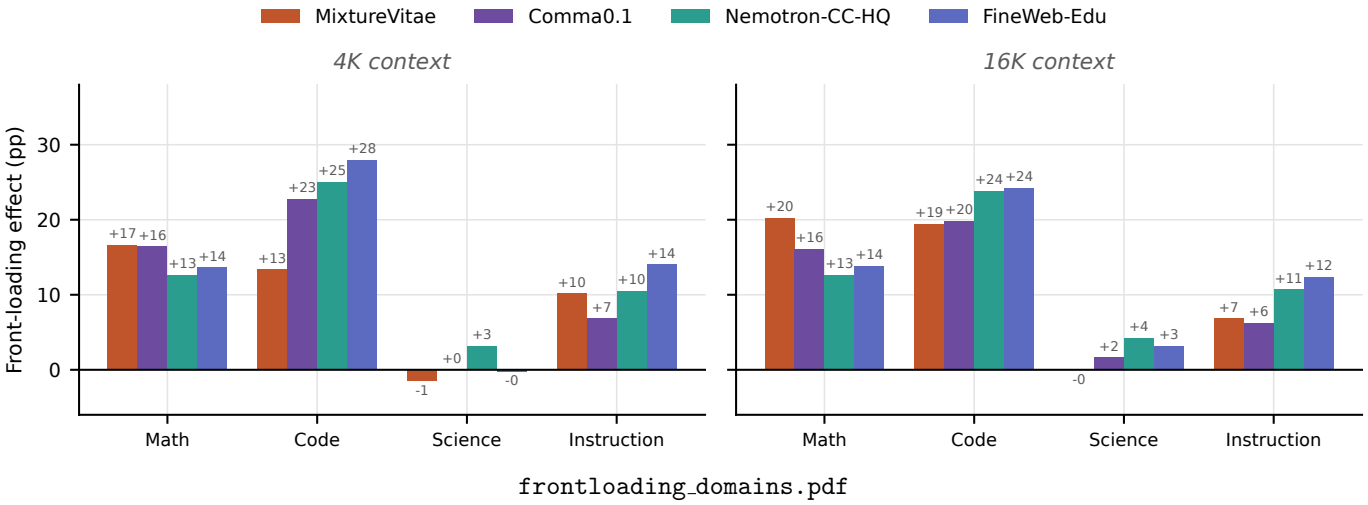


Table 4: Per-benchmark results for the 300B front-loading × source-pool factorial. (FL) marks the models whose pretraining mixture includes the MixtureVitae instruction-and-reasoning subset. Protocol A (Appendix Table **??**). Values are mean$_{\pm SD}$ over decoding seeds 42, 1234, and 2025. Comma0.1 is a second permissive source pool. Its model without front-loading is the Comma0.1 model of Table **??**, and the front-loaded MixtureVitae model is the main MixtureVitae model. Both are reused checkpoints, trained earlier than the other models here with the same recorded `open-sci-ref` configuration. The table has two panels: (a) mathematics, (b) code, science and instruction following, and the suite average.

(a) Mathematics

Model	A24	A25	AMC	M500	GSM8K
MixtureVitae (FL)	$0.0_{\pm 0.00}$	$7.8_{\pm 5.09}$	$5.0_{\pm 0.00}$	$17.8_{\pm 0.72}$	$54.1_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$12.5_{\pm 0.00}$	$29.5_{\pm 0.23}$	$49.9_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$3.3_{\pm 0.00}$	$10.8_{\pm 1.44}$	$30.5_{\pm 0.12}$	$50.2_{\pm 0.00}$
MixtureVitae	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.1_{\pm 0.12}$	$3.3_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$1.9_{\pm 0.12}$	$4.4_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$1.7_{\pm 0.23}$	$4.0_{\pm 0.00}$
Comma0.1 (FL)	$0.0_{\pm 0.00}$	$12.2_{\pm 1.92}$	$21.7_{\pm 1.44}$	$52.9_{\pm 0.31}$	$57.7_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$8.3_{\pm 2.89}$	$29.2_{\pm 1.04}$	$56.5_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$16.7_{\pm 2.89}$	$30.7_{\pm 0.31}$	$56.6_{\pm 0.00}$
Comma0.1	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.5_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.23}$	$9.3_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.31}$	$9.7_{\pm 0.00}$
Nemotron (FL)	$1.1_{\pm 1.92}$	$5.6_{\pm 1.92}$	$25.0_{\pm 0.00}$	$44.6_{\pm 0.69}$	$52.7_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$7.5_{\pm 0.00}$	$19.4_{\pm 0.40}$	$50.9_{\pm 0.00}$
→ SFT → DPO	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$7.5_{\pm 2.50}$	$19.9_{\pm 0.99}$	$51.5_{\pm 0.00}$
Nemotron	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.2_{\pm 0.00}$	$4.1_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.0_{\pm 0.00}$	$3.1_{\pm 0.23}$	$6.8_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$3.5_{\pm 0.42}$	$8.0_{\pm 0.00}$
FineWeb-Edu (FL)	$0.0_{\pm 0.00}$	$6.7_{\pm 0.00}$	$22.5_{\pm 0.00}$	$47.0_{\pm 1.11}$	$53.5_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$9.2_{\pm 1.44}$	$17.9_{\pm 0.42}$	$50.1_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$7.5_{\pm 0.00}$	$18.4_{\pm 0.20}$	$50.2_{\pm 0.00}$
FineWeb-Edu	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$3.2_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.1_{\pm 0.12}$	$7.1_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$1.7_{\pm 0.31}$	$6.7_{\pm 0.00}$
MixtureVitae-16K (FL)	$1.1_{\pm 1.92}$	$3.3_{\pm 0.00}$	$23.3_{\pm 8.04}$	$47.0_{\pm 0.53}$	$60.0_{\pm 0.00}$
→ SFT	$3.3_{\pm 0.00}$	$3.3_{\pm 0.00}$	$15.0_{\pm 0.00}$	$35.0_{\pm 0.53}$	$51.6_{\pm 0.00}$
→ SFT → DPO	$3.3_{\pm 0.00}$	$1.1_{\pm 1.92}$	$17.5_{\pm 0.00}$	$37.0_{\pm 0.80}$	$52.6_{\pm 0.00}$
MixtureVitae-16K	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$2.6_{\pm 0.20}$	$5.4_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.3_{\pm 0.12}$	$5.0_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.8_{\pm 1.44}$	$2.6_{\pm 0.20}$	$4.6_{\pm 0.00}$
Comma0.1-16K (FL)	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$23.3_{\pm 2.89}$	$41.8_{\pm 0.00}$	$59.6_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$10.0_{\pm 0.00}$	$26.7_{\pm 0.31}$	$55.4_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$9.2_{\pm 1.44}$	$27.7_{\pm 0.12}$	$55.9_{\pm 0.00}$
Comma0.1-16K	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 1.44}$	$2.8_{\pm 0.35}$	$12.2_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$2.5_{\pm 0.12}$	$6.9_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$2.1_{\pm 0.12}$	$6.7_{\pm 0.00}$
Nemotron-16K (FL)	$0.0_{\pm 0.00}$	$1.1_{\pm 1.92}$	$20.0_{\pm 2.50}$	$39.9_{\pm 0.50}$	$59.8_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$6.7_{\pm 1.44}$	$18.5_{\pm 0.46}$	$49.6_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.8_{\pm 1.44}$	$20.3_{\pm 0.50}$	$50.7_{\pm 0.00}$
Nemotron-16K	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.8_{\pm 0.35}$	$8.2_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.9_{\pm 0.31}$	$8.6_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$2.5_{\pm 0.12}$	$7.7_{\pm 0.00}$
FineWeb-Edu-16K (FL)	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$13.3_{\pm 1.44}$	$38.0_{\pm 0.35}$	$57.2_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$2.2_{\pm 1.92}$	$5.8_{\pm 1.44}$	$21.9_{\pm 0.64}$	$51.1_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$23.1_{\pm 0.31}$	$51.1_{\pm 0.00}$
FineWeb-Edu-16K	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.0_{\pm 0.00}$	$1.8_{\pm 0.20}$	$8.1_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$3.1_{\pm 0.12}$	$8.7_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$3.3_{\pm 0.00}$	$0.0_{\pm 0.00}$	$3.3_{\pm 0.12}$	$9.2_{\pm 0.00}$

Table 4: (continued) Panel (b), code, science and instruction following, and the suite average.

(b) Code, science and instruction following, and the suite average

Model	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
MixtureVitae (FL)	$13.6_{\pm 0.93}$	$7.7_{\pm 0.16}$	$19.5_{\pm 0.12}$	$24.4_{\pm 0.77}$	$2.9_{\pm 0.51}$	$34.7_{\pm 0.13}$	$17.0_{\pm 0.38}$
→ SFT	$44.3_{\pm 0.70}$	$1.2_{\pm 0.16}$	$39.9_{\pm 0.23}$	$20.7_{\pm 0.87}$	$3.7_{\pm 0.55}$	$48.0_{\pm 0.08}$	$22.7_{\pm 0.14}$
→ SFT → DPO	$47.4_{\pm 0.35}$	$1.0_{\pm 0.16}$	$41.2_{\pm 0.20}$	$21.2_{\pm 0.00}$	$4.5_{\pm 0.34}$	$49.8_{\pm 0.08}$	$23.6_{\pm 0.17}$
MixtureVitae	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$22.0_{\pm 0.20}$	$23.6_{\pm 0.29}$	$0.0_{\pm 0.00}$	$30.9_{\pm 0.33}$	$7.3_{\pm 0.03}$
→ SFT	$19.1_{\pm 0.35}$	$1.1_{\pm 0.00}$	$25.1_{\pm 0.50}$	$25.1_{\pm 0.29}$	$2.3_{\pm 0.30}$	$37.9_{\pm 0.20}$	$10.9_{\pm 0.02}$
→ SFT → DPO	$19.7_{\pm 0.35}$	$1.1_{\pm 0.00}$	$25.4_{\pm 0.00}$	$25.1_{\pm 0.29}$	$2.5_{\pm 0.53}$	$41.7_{\pm 0.15}$	$11.2_{\pm 0.08}$
Comma0.1 (FL)	$44.9_{\pm 0.35}$	$9.4_{\pm 0.63}$	$37.1_{\pm 0.50}$	$24.7_{\pm 2.31}$	$4.3_{\pm 0.26}$	$36.8_{\pm 0.26}$	$27.4_{\pm 0.14}$
→ SFT	$56.7_{\pm 0.00}$	$6.8_{\pm 0.00}$	$42.7_{\pm 0.31}$	$24.2_{\pm 0.00}$	$5.2_{\pm 0.33}$	$49.2_{\pm 0.49}$	$25.4_{\pm 0.27}$
→ SFT → DPO	$57.9_{\pm 0.00}$	$6.7_{\pm 0.31}$	$42.1_{\pm 0.76}$	$25.6_{\pm 0.29}$	$5.2_{\pm 0.33}$	$52.1_{\pm 0.61}$	$26.7_{\pm 0.30}$
Comma0.1	$7.9_{\pm 0.00}$	$0.0_{\pm 0.00}$	$8.8_{\pm 0.20}$	$20.2_{\pm 1.34}$	$0.0_{\pm 0.00}$	$38.4_{\pm 0.00}$	$7.3_{\pm 0.12}$
→ SFT	$15.2_{\pm 0.00}$	$0.2_{\pm 0.16}$	$22.5_{\pm 0.46}$	$25.6_{\pm 0.29}$	$3.9_{\pm 0.25}$	$42.4_{\pm 0.08}$	$11.1_{\pm 0.08}$
→ SFT → DPO	$15.2_{\pm 0.00}$	$0.6_{\pm 0.16}$	$22.7_{\pm 0.12}$	$26.8_{\pm 1.34}$	$3.9_{\pm 0.49}$	$44.7_{\pm 0.53}$	$11.5_{\pm 0.07}$
Nemotron (FL)	$40.2_{\pm 0.61}$	$7.7_{\pm 0.95}$	$8.7_{\pm 0.12}$	$20.4_{\pm 1.27}$	$3.3_{\pm 0.42}$	$33.5_{\pm 0.20}$	$22.1_{\pm 0.22}$
→ SFT	$47.8_{\pm 0.70}$	$2.6_{\pm 0.16}$	$35.1_{\pm 0.12}$	$24.2_{\pm 0.00}$	$4.3_{\pm 0.15}$	$48.5_{\pm 0.53}$	$21.9_{\pm 0.13}$
→ SFT → DPO	$50.0_{\pm 0.00}$	$3.3_{\pm 0.27}$	$35.5_{\pm 0.23}$	$26.1_{\pm 0.29}$	$4.8_{\pm 0.78}$	$51.1_{\pm 0.53}$	$22.8_{\pm 0.25}$
Nemotron	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.4_{\pm 0.00}$	$5.7_{\pm 0.29}$	$0.0_{\pm 0.00}$	$29.2_{\pm 0.08}$	$3.8_{\pm 0.02}$
→ SFT	$4.9_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.5_{\pm 0.23}$	$20.2_{\pm 0.00}$	$2.0_{\pm 0.17}$	$38.1_{\pm 0.33}$	$7.8_{\pm 0.04}$
→ SFT → DPO	$6.1_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.1_{\pm 0.12}$	$21.2_{\pm 0.00}$	$2.7_{\pm 0.66}$	$39.3_{\pm 0.40}$	$8.0_{\pm 0.07}$
FineWeb-Edu (FL)	$38.4_{\pm 0.00}$	$9.8_{\pm 0.54}$	$28.5_{\pm 0.64}$	$23.1_{\pm 0.29}$	$5.0_{\pm 0.39}$	$33.9_{\pm 0.45}$	$24.4_{\pm 0.05}$
→ SFT	$51.2_{\pm 0.00}$	$2.9_{\pm 0.16}$	$34.4_{\pm 0.20}$	$23.6_{\pm 0.29}$	$4.4_{\pm 0.14}$	$48.1_{\pm 0.42}$	$22.0_{\pm 0.14}$
→ SFT → DPO	$53.0_{\pm 0.00}$	$4.1_{\pm 0.00}$	$35.2_{\pm 0.00}$	$23.7_{\pm 0.00}$	$4.4_{\pm 0.18}$	$49.5_{\pm 0.53}$	$22.4_{\pm 0.08}$
FineWeb-Edu	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$25.1_{\pm 0.58}$	$0.0_{\pm 0.00}$	$34.7_{\pm 0.00}$	$5.7_{\pm 0.05}$
→ SFT	$2.4_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.3_{\pm 0.23}$	$26.9_{\pm 0.58}$	$1.6_{\pm 0.46}$	$34.1_{\pm 0.00}$	$6.9_{\pm 0.08}$
→ SFT → DPO	$2.4_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.9_{\pm 0.12}$	$25.9_{\pm 0.29}$	$1.6_{\pm 0.17}$	$33.9_{\pm 0.45}$	$7.0_{\pm 0.06}$
MixtureVitae-16K (FL)	$54.1_{\pm 0.35}$	$11.5_{\pm 0.16}$	$40.1_{\pm 0.23}$	$23.4_{\pm 0.29}$	$8.0_{\pm 0.24}$	$47.7_{\pm 0.34}$	$29.1_{\pm 0.69}$
→ SFT	$54.1_{\pm 0.35}$	$9.7_{\pm 0.16}$	$40.4_{\pm 0.35}$	$23.4_{\pm 0.29}$	$5.1_{\pm 0.27}$	$51.9_{\pm 0.46}$	$26.6_{\pm 0.11}$
→ SFT → DPO	$52.6_{\pm 0.70}$	$9.4_{\pm 0.31}$	$40.3_{\pm 0.12}$	$23.4_{\pm 0.29}$	$6.2_{\pm 0.20}$	$52.6_{\pm 0.60}$	$26.9_{\pm 0.04}$
MixtureVitae-16K	$19.3_{\pm 0.35}$	$0.0_{\pm 0.00}$	$22.3_{\pm 0.23}$	$23.2_{\pm 0.00}$	$6.5_{\pm 0.07}$	$42.7_{\pm 0.08}$	$11.3_{\pm 0.03}$
→ SFT	$19.5_{\pm 0.00}$	$0.5_{\pm 0.00}$	$25.9_{\pm 0.42}$	$25.1_{\pm 0.29}$	$3.4_{\pm 0.18}$	$45.1_{\pm 0.23}$	$11.5_{\pm 0.02}$
→ SFT → DPO	$19.1_{\pm 0.35}$	$0.4_{\pm 0.16}$	$25.6_{\pm 0.00}$	$25.1_{\pm 0.58}$	$3.9_{\pm 0.53}$	$46.4_{\pm 0.23}$	$11.7_{\pm 0.13}$
Comma0.1-16K (FL)	$45.3_{\pm 0.35}$	$2.4_{\pm 0.27}$	$41.4_{\pm 0.20}$	$24.2_{\pm 0.00}$	$10.8_{\pm 1.26}$	$52.8_{\pm 0.20}$	$27.5_{\pm 0.34}$
→ SFT	$54.3_{\pm 0.00}$	$6.1_{\pm 0.16}$	$40.8_{\pm 0.40}$	$24.1_{\pm 0.29}$	$7.2_{\pm 0.83}$	$53.4_{\pm 0.34}$	$25.3_{\pm 0.13}$
→ SFT → DPO	$54.9_{\pm 0.61}$	$6.1_{\pm 0.31}$	$41.8_{\pm 0.20}$	$23.9_{\pm 0.77}$	$6.5_{\pm 0.51}$	$53.4_{\pm 0.65}$	$25.4_{\pm 0.27}$
Comma0.1-16K	$13.4_{\pm 0.00}$	$0.0_{\pm 0.00}$	$16.0_{\pm 0.20}$	$25.8_{\pm 1.01}$	$7.8_{\pm 0.19}$	$41.9_{\pm 0.64}$	$11.1_{\pm 0.22}$
→ SFT	$16.5_{\pm 0.00}$	$0.2_{\pm 0.31}$	$25.2_{\pm 0.35}$	$23.2_{\pm 0.00}$	$4.8_{\pm 0.66}$	$47.2_{\pm 0.08}$	$11.7_{\pm 0.04}$
→ SFT → DPO	$17.1_{\pm 0.00}$	$0.0_{\pm 0.00}$	$25.9_{\pm 0.31}$	$23.4_{\pm 0.29}$	$5.1_{\pm 0.12}$	$48.8_{\pm 0.68}$	$12.0_{\pm 0.04}$
Nemotron-16K (FL)	$46.5_{\pm 0.35}$	$6.7_{\pm 0.16}$	$34.7_{\pm 0.12}$	$27.3_{\pm 0.00}$	$8.3_{\pm 0.83}$	$46.8_{\pm 0.15}$	$26.5_{\pm 0.39}$
→ SFT	$49.8_{\pm 0.35}$	$0.3_{\pm 0.00}$	$32.5_{\pm 0.23}$	$25.9_{\pm 0.29}$	$6.0_{\pm 0.29}$	$53.6_{\pm 0.34}$	$22.1_{\pm 0.27}$
→ SFT → DPO	$49.4_{\pm 0.61}$	$0.3_{\pm 0.00}$	$32.9_{\pm 0.12}$	$24.7_{\pm 0.00}$	$6.8_{\pm 0.50}$	$55.4_{\pm 0.72}$	$22.4_{\pm 0.26}$
Nemotron-16K	$1.8_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.7_{\pm 0.31}$	$22.7_{\pm 0.51}$	$5.6_{\pm 1.04}$	$39.0_{\pm 0.13}$	$7.8_{\pm 0.07}$
→ SFT	$5.3_{\pm 0.35}$	$0.0_{\pm 0.00}$	$5.9_{\pm 0.12}$	$22.2_{\pm 0.87}$	$1.2_{\pm 0.08}$	$42.9_{\pm 0.13}$	$8.1_{\pm 0.11}$
→ SFT → DPO	$5.3_{\pm 0.35}$	$0.0_{\pm 0.00}$	$5.3_{\pm 0.12}$	$23.4_{\pm 1.17}$	$1.7_{\pm 0.37}$	$43.3_{\pm 0.42}$	$8.3_{\pm 0.06}$
FineWeb-Edu-16K (FL)	$43.9_{\pm 0.00}$	$8.2_{\pm 0.31}$	$35.3_{\pm 0.12}$	$29.0_{\pm 0.29}$	$9.1_{\pm 0.97}$	$52.4_{\pm 0.34}$	$26.0_{\pm 0.06}$
→ SFT	$45.5_{\pm 0.35}$	$2.7_{\pm 0.00}$	$31.7_{\pm 0.23}$	$26.8_{\pm 0.51}$	$5.4_{\pm 0.30}$	$51.5_{\pm 0.08}$	$22.2_{\pm 0.25}$
→ SFT → DPO	$45.1_{\pm 0.61}$	$2.6_{\pm 0.16}$	$31.4_{\pm 0.00}$	$25.3_{\pm 0.00}$	$5.1_{\pm 1.05}$	$53.7_{\pm 0.00}$	$21.8_{\pm 0.07}$
FineWeb-Edu-16K	$3.7_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 0.23}$	$25.4_{\pm 0.29}$	$7.9_{\pm 1.02}$	$38.1_{\pm 0.65}$	$8.3_{\pm 0.10}$
→ SFT	$4.9_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.23}$	$23.6_{\pm 0.58}$	$2.4_{\pm 0.40}$	$39.2_{\pm 0.27}$	$7.7_{\pm 0.07}$
→ SFT → DPO	$4.9_{\pm 0.00}$	$0.1_{\pm 0.16}$	$2.5_{\pm 0.12}$	$23.9_{\pm 0.29}$	$2.7_{\pm 0.54}$	$40.7_{\pm 0.53}$	$8.2_{\pm 0.07}$

Table 5: Language understanding for the sixteen factorial checkpoints, on the 4K pretrained bases and their 16K extended checkpoints. This suite is scored by the LM Evaluation Harness under its own settings and is not on either reasoning protocol. Single seed. Accuracies in percent, `acc_norm` where the harness reports it, and shot counts are in Table **??**. Without front-loading, FineWeb-Edu and the MixtureVitae source pool sit at chance on MMLU and CommonsenseQA. Comma0.1 without front-loading reuses the Comma0.1 model of Table **??**, and it also sits at chance on MMLU and CommonsenseQA. The front-loaded MixtureVitae rows reuse the main MixtureVitae model, as in Table **??**. Table **??** gives 95% intervals for its contrasts.

Model	FL	MMLU	HellaS	CSQA	ARC-C	ARC-E	PIQA	BoolQ	WinoG	OBQA	COPA	LAMBADA	Avg
MixtureVitae	yes	41.9	54.5	51.4	40.3	70.9	70.4	73.5	57.8	34.2	71.0	48.1	55.8
MixtureVitae	no	25.8	52.2	22.4	38.2	68.8	70.5	64.8	58.2	36.4	73.0	51.2	51.0
Comma0.1	yes	40.4	55.4	52.1	39.8	69.4	71.2	75.3	59.5	36.2	70.0	53.4	56.6
Comma0.1	no	26.5	53.5	20.4	36.2	63.1	71.2	63.2	59.3	32.6	70.0	54.5	50.0
Nemotron	yes	50.5	68.9	62.1	52.3	80.3	77.6	79.3	62.9	38.6	80.0	57.7	64.6
Nemotron	no	46.4	71.5	48.6	52.4	80.9	78.7	77.4	65.0	42.0	85.0	58.0	64.2
FineWeb-Edu	yes	47.5	65.3	55.7	49.5	79.9	76.9	77.3	62.6	43.0	79.0	54.6	62.8
FineWeb-Edu	no	26.1	67.1	21.1	48.5	76.9	77.1	64.9	62.4	42.4	82.0	55.7	56.7
MixtureVitae-16K	yes	41.1	53.1	48.8	41.6	71.4	69.7	72.6	57.9	31.8	67.0	49.4	55.0
MixtureVitae-16K	no	26.1	51.0	22.5	38.9	68.6	69.9	61.9	57.9	34.6	76.0	51.8	50.8
Comma0.1-16K	yes	40.4	53.7	49.0	41.9	71.9	70.7	78.3	59.4	35.6	73.0	53.6	57.0
Comma0.1-16K	no	26.8	52.5	22.8	39.7	68.1	71.8	63.9	60.9	33.4	76.0	57.2	52.1
Nemotron-16K	yes	49.5	67.7	61.7	52.3	81.4	77.9	79.5	62.8	38.2	81.0	58.5	64.6
Nemotron-16K	no	46.5	70.0	50.6	52.0	80.7	78.1	77.1	64.6	41.8	83.0	60.1	64.1
FineWeb-Edu-16K	yes	46.4	62.5	56.4	50.2	79.6	76.7	77.6	62.1	41.8	79.0	56.1	62.6
FineWeb-Edu-16K	no	26.8	64.7	20.9	48.1	78.8	76.6	70.0	63.1	42.8	81.0	57.5	57.3